GPS-CLIP: GMM-Based Pseudo-Label Refinement for Semi-Supervised Zero-Shot Remote Sensing Classification

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***Abstract*—Remote sensing scene classification (RSSC) plays a crucial role in applications, such as environmental monitoring, urban planning, and disaster management. However, traditional methods often require large amounts of labeled data, which can be both costly and time-consuming to obtain, especially for rare or unseen categories. Zero-shot RSSC (ZSRSSC) aims to overcome this challenge by recognizing categories that were not present in the training set, leveraging external semantic information. Recent advances in vision–language models (VLMs), such as contrastive language–image pretraining (CLIP), have made it possible to align image features with textual descriptions to tackle this problem. We introduce GPS-CLIP, a novel Gaussian mixture model (GMM)-based framework that leverages Pseudo-label refinement for Semi-supervised learning with CLIP, enabling effective adaptation to the target domain without any labeled target-domain samples. Under an inductive protocol, adaptation uses only the training split; the frozen model is evaluated on a disjoint test split that never participates in adaptation. GPS-CLIP first generates initial pseudo-labels using a pretrained CLIP model, which are then refined through a GMM that captures feature distribution and label structure. Contrastive fine-tuning progressively adapts the model using balanced, high-confidence samples. By integrating these modules into a unified pipeline, GPS-CLIP improves pseudo-label reliability and visual-semantic alignment. Across five benchmarks and three backbones, GPS-CLIP delivers superior or competitive performance under inductive evaluation.**

***Index Terms*—Pseudo-label, remote sensing scene classification (RSSC), semi-supervised learning, zero-shot learning (ZSL).**

I. Introduction

REMOTE sensing scene classification (RSSC) plays a crucial role in a wide range of applications, such as urban planning, environmental monitoring [1], [2], [3], [4], disaster management [5], and agricultural monitoring. In these diverse domains, the ability to automatically categorize remote sensing images directly supports decision-making processes and improves operational efficiency. These tasks often involve the analysis of large geographical areas, where accurate classification of scene types, land use, and natural resources is essential.

However, traditional RSSC methods heavily rely on large amounts of labeled data [6], [7], which is both time-consuming and labor-intensive to acquire, and is also costly. This dependency significantly limits the scalability of such methods, especially when applied to rapidly changing environments where labeled samples are scarce. Furthermore, these methods often struggle with generalization when dealing with unseen scene classes that were not part of the training set [8], [9]. Zero-shot learning (ZSL) techniques offer a promising solution to address these challenges. By bypassing the strict reliance on annotated training data, ZSL shifts the focus toward exploiting semantic relationships between classes. By enabling models to recognize unseen categories using auxiliary semantic information, such as text descriptions or attribute vectors [10], [11], [12], ZSL enhances model adaptability, which is especially important in real-world scenarios where new and evolving scene types are often encountered. In the context of remote sensing, zero-shot RSSC (ZSRSSC) leverages this approach to enable the classification of previously unseen classes without requiring additional labeled data.

Building upon the foundations of ZSL, the emergence of large-scale pretrained multimodal models has further broadened the horizon for cross-domain applications. Recent advances in vision–language models (VLMs) have significantly contributed to ZSL tasks [13], [14], enabling the alignment of visual data with textual information. Among these models, some have become particularly influential due

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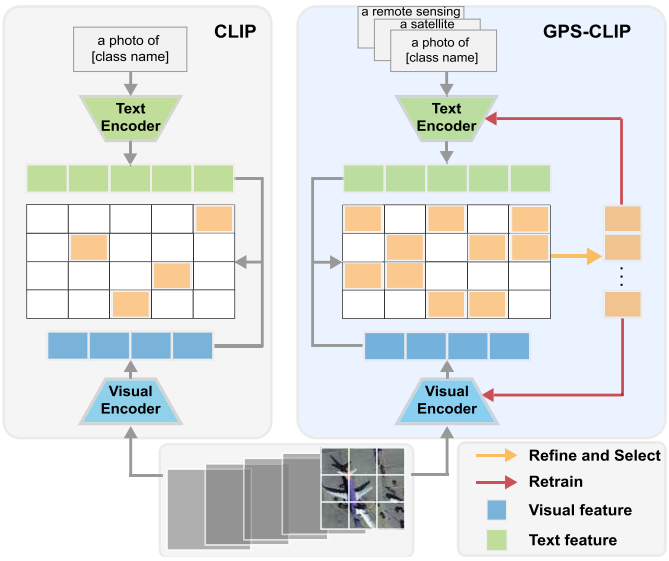


Fig. 1. Comparison of CLIP and GPS-CLIP. (Left) CLIP zero-shot pipeline, where image and text encoders map visual inputs and natural language descriptions into a shared space. (Right) GPS-CLIP, combining pseudo-label refinement and progressive sample selection in a semi-supervised learning framework.

to their remarkable transferability across tasks. These models bridge the gap between image content and natural language descriptions by jointly learning visual and semantic representations [15], [16], [17], [18], which allows for more flexible and robust recognition of novel categories. A notable example is the contrastive language–image pretraining (CLIP) model [19], which has demonstrated impressive zero-shot transfer capabilities in various computer vision tasks, including object detection and classification. However, directly applying these general VLMs to RSSC still faces several challenges. Remote sensing images exhibit unique characteristics, such as large-scale scenes, complex backgrounds, and diverse object distributions, which are quite different from the natural images typically used in the pretraining of general VLMs. These differences complicate the application of standard VLMs to remote sensing data, highlighting the need for specialized techniques that can better handle the specific challenges of remote sensing imagery.

Motivated by the limitations of existing approaches, our framework seeks to combine the strengths of zero-shot generalization with domain-specific refinements. To address the domain adaptation issue of CLIP in remote sensing, we propose a novel method called GPS-CLIP to enhance the classification performance of ZSRSSC. Compared to the existing CLIP model, we introduce a pseudo-label-based semi-supervised learning framework and a pseudo-label refinement module based on Gaussian clustering. As shown in Fig. 1, the left side illustrates the ZSL pipeline of CLIP, where the image encoder and text encoder are jointly trained to map visual inputs and natural language descriptions into a shared embedding space. During inference, the name of each candidate class is first converted into a simple prompt, such as 'a photograph of [class]', and encoded into a textual feature, while the input image is processed into a visual feature. In contrast, on the right side of Fig. 1, we first adopt multiple natural language descriptions and calculate their average value. Subsequently, we feed the preliminary results obtained from the inference of the pretrained CLIP model into the pseudo-label refinement module, while implementing a progressive selection strategy during this process (as indicated by the orange arrow). This module models each sample's probability distribution using a Gaussian mixture model (GMM) [20], thus generating more accurate pseudo-labels. Additionally, within the pseudo-label-based semi-supervised learning framework, we employ a progressive sample selection strategy, indicated by the red arrow, which gradually increases the number of high-confidence pseudo-labeled samples from the unlabeled training split. Compared with the existing work, our proposed GPS-CLIP makes the following contributions.

1) We adapt GMM-based assignment refinement to an inductive remote-sensing protocol by estimating class-conditional density, neighborhood relations, and semantic consistency exclusively from the unlabeled training split.
2) We develop a class-balanced progressive selection strategy that uses refined pseudo-labels to adapt the CLIP projection layers to the target domain.
3) Extensive experiments demonstrate superior or competitive performance on five benchmark datasets. Controlled ablations keep the inherited prompt ensemble fixed, isolate the effect of GMM refinement, and examine progressive-schedule sensitivity.

II. Related Work

A. Zero-Shot RSSC

ZSL offers a framework for recognizing novel categories by leveraging semantic information. This paradigm is particularly attractive in scenarios where new scene categories emerge frequently, yet collecting labeled data remains infeasible. Classical methods typically map visual features into semantic spaces based on attributes [10], [11], [12], [21] or word embeddings [8], [22], [23]. By projecting images into these auxiliary semantic domains, models can bridge the gap between seen and unseen categories, thereby facilitating generalization beyond the training set. These include approaches like the direct attribute prediction (DAP) model and compatibility functions for semantic alignment. Their pioneering work laid the foundation for later studies that sought to combine handcrafted representations with semantic embeddings, gradually moving toward the zero-shot setting. In the context of remote sensing, Yang and Newsam [24] initiated research on visual representation for land-use classification using Bag-of-Visual-Words (BoVWs), which later influenced zero-shot formulations.

With the advent of large pretrained multimodal models, the community began exploring ways to inject remote sensing knowledge into these general-purpose frameworks. Li et al. [25] introduced RS-CLIP, one of the first attempts to adapt large-scale VLMs for remote sensing. Their framework integrates a contrastive learning objective between image and text representations, supported by a pseudo-labeling strategy

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and curriculum learning to iteratively refine supervision. Quan et al. [26] further enhanced ZSL in RS through Sammon embedding and manifold regularization, which helps preserve class-wise semantic structure. To address limitations in data diversity, Li et al. [27] proposed CSPWGAN to synthesize semantic features for unseen classes using GANs, building on generative approaches like CADA-VAE [28].

In addition, Cheng et al. [5] conducted a comprehensive review of scene classification datasets and methods, identifying the need for generalizable approaches in data-scarce remote sensing domains. Their findings emphasize that without mechanisms for knowledge transfer, models trained on one dataset often fail when applied to new geographic regions or different imaging conditions. Their study underlines the domain shift challenge where high intraclass variance and background noise limit traditional models [5], [29]. To this end, modern approaches increasingly rely on richer text semantics, large-scale pretraining, and alignment with downstream tasks to improve generalization across unseen categories [30], [31], [32]. These trends collectively signal a paradigm shift from handcrafted features toward foundation-style models that are capable of handling the diversity of RS imagery.

B. VLMs in Remote Sensing

The success of CLIP [19] has popularized vision–language contrastive models capable of transferring semantic understanding from natural language supervision to a broad range of visual tasks. As one of the most influential representatives, CLIP demonstrated that large-scale pretraining with noisy internet data could yield surprisingly strong transfer performance. These models jointly learn a shared embedding space for images and text using large-scale image–text pairs, enabling impressive zero-shot performance. However, remote sensing imagery presents unique challenges, such as scale variation, lack of contextual cues, and domain gap from natural images.

GPS-CLIP jointly considers class-conditional Gaussian density, neighborhood affinity, and semantic consistency. Unlike label propagation, it explicitly models class-conditional feature distributions; unlike conventional GMM refinement, it also preserves neighborhood and semantic constraints. The GMM-based assignment refinement, including Laplacian and KL regularization and its MM optimization, follows Zanella et al. [20] and is not claimed as an independently new formulation. Our contribution is to use this refinement only on an unlabeled training split, convert the assignments into class-balanced pseudo-supervision, progressively update the CLIP projection layers, and then freeze the model for unseen test inference.

Dataset construction plays a pivotal role here, since the quality and diversity of image–text pairs directly affect the transferability of pretrained models. Other notable works, such as RemoteCLIP [33] and GeoRSCLIP [34], extend the VLM framework to RS by curating large-scale datasets with geospatial image–text pairs. These models emphasize domain-specific vocabulary and visual priors, which significantly improve performance in retrieval, captioning, and classification tasks. Moving beyond task-specific adaptations, researchers have also begun exploring general-purpose RS foundation models that integrate multiple modalities. More recently, RingmoGPT [35] aims to unify vision, language, and spatial grounding in a single RS foundation model, outperforming prior CLIP-based variants in multitask learning scenarios. As highlighted in [36], effective prompt engineering and adaptation strategies play a key role in transferring general-purpose language models to the RS domain [13], [14]. Bang et al. [37] further extend this by introducing active prompt selection to enhance label efficiency and adaptiveness in VLMs.

C. Semi-Supervised Contrastive Learning via Pseudo-Labels

Pseudo-labeling [38] is a widely used semi-supervised learning strategy in which a model's high-confidence predictions on unlabeled data are treated as surrogate labels to augment the supervised training set. This approach enables the model to leverage large amounts of unlabeled data, which is particularly useful when annotated samples are limited or costly to obtain. However, pseudo-labeling may inadvertently reinforce incorrect predictions, especially in the presence of class imbalance or noisy predictions. Errors introduced in the early stages of training can propagate and degrade overall performance, highlighting the need for more robust integration strategies.

To address this issue, Cascante-Bonilla et al. [39] propose a curriculum-based approach that progressively incorporates pseudo-labeled examples according to their confidence scores. By introducing samples in ascending order of confidence, the model is less likely to overfit noisy labels and can gradually adapt to the structure of the unlabeled data. Such curriculum-driven strategies are particularly advantageous in remote sensing tasks, where labeled datasets are often small and heterogeneous. In this context, a pretrained CLIP model can be leveraged to generate pseudo-labels by comparing the image embeddings to a set of textual class prompts. These pseudo-labeled examples can then be integrated into the contrastive learning process to refine alignment between visual and textual representations [40], enabling CLIP to adapt to novel target classes without requiring any manually annotated data. Recent works further enhance robustness by combining pseudo-labeling with contrastive objectives, encouraging the model to learn more discriminative features from both labeled and unlabeled samples.

III. Methodology

A. Overall Framework

GPS-CLIP is an enhanced inductive zero-shot classification framework designed for remote sensing scenes. As shown in Fig. 2, it first uses CLIP's visual encoder to extract features from each image in the unlabeled training split and its text encoder to embed each class name. Temperature-scaled cosine similarity then generates initial pseudo-labels. In this stage, the visual features build the affinity matrix ω_{ij} and are processed by a GMM to yield assignment probabilities $p_{i,k}$.

Then, these initial pseudo-labels are refined via a direct GMM. We alternate between decoupled MM updates of the assignments and closed-form updates of class mean vectors μ_k and a shared covariance matrix Σ. To address class imbalance,

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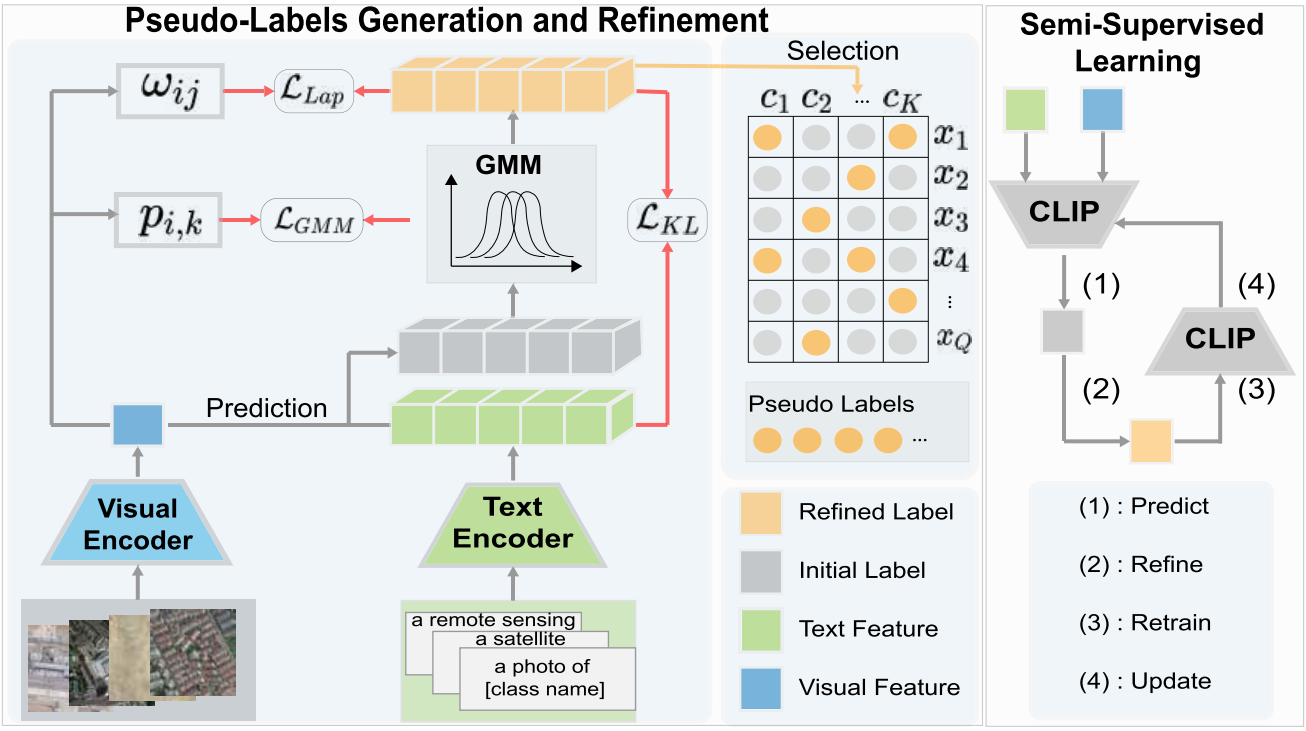


Fig. 2. Framework of GPS-CLIP. (Left) (pseudo-label generation), CLIP's visual and text encoders extract features and class embeddings; initial soft pseudo-labels are generated, the affinity matrix ω_{ij} is constructed, and probabilities $p_{i,k}$ are obtained. The pseudo-labels are then refined using a GMM and subjected to sample selection. (Right) (semi-supervised learning), CLIP is fine-tuned on the growing pseudo-labeled dataset.

we select an equal number of high-confidence samples per class after optimization. We ensure that the number of newly added samples increases steadily with each iteration.

In the end, we fine-tune the CLIP model over multiple rounds on this progressively expanding, balanced pseudo-labeled dataset. As sample confidence improves, the fine-tuning process leverages increasingly reliable data, enabling the model to adapt to and recognize unseen target classes without any ground-truth labels.

B. Pseudo-Labels Generation

Let $\mathcal{D}_u^{\text{tr}} = \{\mathbf{x}_i\}_{i=1}^{Q}$ denote the unlabeled training split of Q images and $\{c_k\}_{k=1}^{K}$ the corresponding set of K class name prompts. We employ a pretrained CLIP model to obtain representations: each image $\mathbf{x}_i$ is encoded by the visual encoder f into an embedding $v_i = f(\mathbf{x}_i)$, and each class name prompt c_k is encoded by the text encoder g into $t_k = g(c_k)$.

For every image–class pair $(\mathbf{x}_i, c_k)$, we compute the cosine similarity between v_i and t_k, scale it by a temperature τ, and normalize across all K classes via softmax

$$\hat{y}_{i,k} = \frac{\exp\left(v_i^\top t_k/\tau\right)}{\sum_{j=1}^{K}\exp\left(v_i^\top t_j/\tau\right)}. \tag{1}$$

The resulting vector $\hat{\mathbf{y}}_i = (\hat{y}_{i,1}, \ldots, \hat{y}_{i,K}) \in \Delta_K$ serves as the initial pseudo-label for image $\mathbf{x}_i$, where Δ_K denotes the K-dimensional probability simplex.

C. Pseudo-Labels Refinement

To improve the quality of pseudo-labels, we adapt the assignment refinement of Zanella et al. [20] to the unlabeled training split. Specifically, we represent the distribution of image embeddings v_i as a class-aligned mixture of K Gaussian components, with one component corresponding to each class. The mixture is parameterized by a shared diagonal covariance Σ and per-class means μ_k. We initialize each μ_k by averaging the embeddings of the eight most confident samples under the initial pseudo-labels, and set every diagonal entry of Σ to $1/d$. We use a shared diagonal covariance to obtain stable estimates from a small number of high-confidence initial samples. This design reduces the risk of ill-conditioned class-specific covariance estimates when training labels are unavailable. All Gaussian statistics are estimated only from $\mathcal{D}_u^{\text{tr}}$. Under this GMM formulation, the likelihood that v_i belongs to component k is given by

$$p_{i,k} \propto \det(\Sigma)^{-\frac{1}{2}} \exp\left(-\frac{1}{2}(v_i - \mu_k)^\top \Sigma^{-1}(v_i - \mu_k)\right) \tag{2}$$

where $p_{i,k}$ denotes the probability of assigning embedding v_i to the kth Gaussian component.

1) Objective Function: We then optimize a composite objective that combines three terms: GMM clustering $\mathcal{L}_{\text{GMM}}$, Laplacian regularization $\mathcal{L}_{\text{Lap}}$, and KL regularization $\mathcal{L}_{\text{KL}}$. Let $\mathbf{z}_i = (z_{i,1}, \ldots, z_{i,K}) \in \Delta_K$ be the refined soft assignment for sample i, and let Q index only the samples in $\mathcal{D}_u^{\text{tr}}$.

GMM clustering encourages the predicted assignment $\mathbf{z}_i$ to align with the distributional likelihood of the sample under a Gaussian mixture model. The GMM loss is then expressed as

$$\mathcal{L}_{\text{GMM}} = -\frac{1}{|Q|}\sum_{i\in Q}\mathbf{z}_i^\top \log(p_i). \tag{3}$$

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Laplacian regularization encourages similar samples to have similar assignments by incorporating affinity structure among image embeddings. Specifically, we define the affinity weight $w_{ij} = \max(0, v_i^\top v_j)$ as the cosine similarity between embeddings v_i and v_j. The loss takes the form

$$\mathcal{L}_{\text{Lap}} = -\sum_{i\in Q}\sum_{j\in Q} w_{ij}\,\mathbf{z}_i^\top \mathbf{z}_j. \tag{4}$$

This formulation promotes smoothness in the label space and leverages the underlying geometry of the embedding manifold. Note that $\mathcal{L}_{\text{Lap}}$ carries a negative sign: minimizing $\mathcal{L}_{\text{Lap}}$ is equivalent to maximizing $\sum_{i,j} w_{ij}\,\mathbf{z}_i^\top \mathbf{z}_j$, which encourages samples with high affinity (large w_{ij}) to share similar soft assignments—consistent with the graph smoothness principle. The update rule in (7) is derived from the multiplicative MM update that jointly optimizes $\mathcal{L}_{\text{GMM}}$, $\mathcal{L}_{\text{Lap}}$, and $\mathcal{L}_{\text{KL}}$ with respect to $\mathbf{z}_i$ on the probability simplex Δ_K.

KL regularization prevents the assignment variable $\mathbf{z}_i$ from deviating too far from the initial pseudo-label $\hat{\mathbf{y}}_i$. The KL regularization loss is then defined as

$$\mathcal{L}_{\text{KL}} = \sum_{i\in Q} \text{KL}\left(\mathbf{z}_i \,\|\, \hat{\mathbf{y}}_i\right) \tag{5}$$

which integrates text-based knowledge and anchors the optimization to semantically meaningful assignments.

Finally, the total refinement loss $\mathcal{L}$ is

$$\mathcal{L}\left(\mathbf{Z}, \boldsymbol{\mu}, \boldsymbol{\Sigma}\right) = \mathcal{L}_{\text{GMM}} + \mathcal{L}_{\text{Lap}} + \mathcal{L}_{\text{KL}} \tag{6}$$

where $\mathbf{Z} = (\mathbf{z}_1, \ldots, \mathbf{z}_Q)$ and $\boldsymbol{\mu} = (\mu_1, \ldots, \mu_K)$.

2) Solving Procedure: First, all image and text embeddings are computed once. Then, after constructing the affinity matrix w_{ij} and obtaining the initial pseudo-labels $\hat{\mathbf{y}}_i$, we initialize the assignment vectors $\mathbf{z}_i^{(1)} = \hat{\mathbf{y}}_i$ and the GMM parameters μ_k, Σ. The algorithm proceeds by alternating between iterative decoupled updates—which update each assignment $\mathbf{z}_i$ in parallel according to its neighbors and closed-form updates for the GMM parameters.

In the iterative decoupled update at iteration l, each assignment vector $\mathbf{z}_i$ is updated as

$$\mathbf{z}_i^{(l+1)} = \frac{\hat{\mathbf{y}}_i \;\odot\; \exp\left(\log p_i \;+\; \sum_{j\in Q} w_{ij}\,\mathbf{z}_j^{(l)}\right)}{\left(\hat{\mathbf{y}}_i \;\odot\; \exp\left(\log p_i + \sum_{j\in Q} w_{ij}\,\mathbf{z}_j^{(l)}\right)\right)^\top \mathbf{1}_K}. \tag{7}$$

Once the assignment vectors $\mathbf{z}_i$ have been updated, the Gaussian mixture parameters admit closed-form solutions: the class mean μ_k and shared covariance diagonal of Σ are given by

$$\mu_k = \frac{\sum_{i\in Q} \mathbf{z}_{i,k}\, v_i}{\sum_{i\in Q} \mathbf{z}_{i,k}} \tag{8}$$

$$\text{diag}\left(\Sigma\right) = \frac{1}{|Q|}\sum_{i\in Q}\sum_{k=1}^{K} \mathbf{z}_{i,k}\,\left(v_i - \mu_k\right)^2. \tag{9}$$

These two update steps are alternated until convergence of the assignment variables $\mathbf{z}_i$ and the GMM parameters μ_k, Σ.

D. Semi-Supervised Contrastive Learning

To better guide the CLIP model, we select an equal number of pseudo-labeled samples for each class, which helps prevent class imbalance issues during pseudo-label selection. Specifically, for each class c, we select the top N samples with the highest confidence scores from the prediction matrix $\mathbf{Z}$, denoted as

$$\mathbf{s}_c = \text{Top-}N\left(\mathbf{Z}_c\right) \tag{10}$$

where $\mathbf{Z}_c$ denotes the cth column of probability matrix $\mathbf{Z}$. Finally, the complete set of pseudo-labeled samples is represented as

$$\mathbf{S} = \bigcup \{\mathbf{s}_c\}_{c=1}^{K}. \tag{11}$$

Although the pseudo-labels obtained after refining are relatively accurate, selecting too many samples at once may introduce low-confidence predictions and degrade classification performance. To mitigate this, we progressively incorporate more reliable unlabeled samples into training. Specifically, in the early stages, the model is less sensitive to remote sensing images, so we only select a small number of high-confidence samples as pseudo-labels. As the model gradually adapts to the target domain, more samples are included. At the rth iteration, we select N_r samples as pseudo-labels, where $N_r \geq N_{r-1}$, ensuring that the number of pseudo-labeled samples selected in each round does not decrease.

Finally, we use the selected pseudo-labeled set S to fine-tune the CLIP model by optimizing its standard contrastive loss. Concretely, consider a mini-batch of B pseudo-labeled samples $\{(\mathbf{x}_i, \mathbf{z}_i)\}_{i=1}^{B} \subset S$, where $\mathbf{z}_i \in \{1, \ldots, K\}$ is the pseudo-label for image $\mathbf{x}_i$. We encode $\mathbf{x}_i$ to obtain $v_i = f(\mathbf{x}_i)$ and get the corresponding class text embedding $t_{\mathbf{z}_i} = g(c_{\mathbf{z}_i})$, where $c_{\mathbf{z}_i}$ is the class name prompt.

The CLIP contrastive loss is defined as the average of two symmetric cross-modal objectives

$$\mathcal{L}_{\text{CLIP}}^{\text{image}} = -\frac{1}{B}\sum_{i=1}^{B} \log \frac{\exp\left(v_i^\top t_{\mathbf{z}_i}/\tau\right)}{\sum_{j=1}^{B} \exp\left(v_i^\top t_{\mathbf{z}_j}/\tau\right)} \tag{12}$$

$$\mathcal{L}_{\text{CLIP}}^{\text{text}} = -\frac{1}{B}\sum_{i=1}^{B} \log \frac{\exp\left(v_i^\top t_{\mathbf{z}_i}/\tau\right)}{\sum_{j=1}^{B} \exp\left(v_j^\top t_{\mathbf{z}_i}/\tau\right)} \tag{13}$$

where τ is a temperature hyperparameter. The final loss is

$$\mathcal{L}_{\text{CLIP}} = \mathcal{L}_{\text{CLIP}}^{\text{image}} + \mathcal{L}_{\text{CLIP}}^{\text{text}}. \tag{14}$$

This objective encourages image features to align with the correct class text features while being distinguishable from other classes. By minimizing $\mathcal{L}_{\text{CLIP}}$ on the pseudo-labeled set $S \subset \mathcal{D}_u^{\text{tr}}$, the CLIP model is further adapted to the target domain.

E. GPS-CLIP Training Workflow

To better illustrate the overall model algorithm, the training procedure is summarized in Algorithm 1. In each outer iteration, we first extract visual features $v_i = f(\mathbf{x}_i)$ from the unlabeled training images $\mathbf{x}_i \in \mathcal{D}_u^{\text{tr}}$ using the CLIP image encoder f, and textual features $t_k = g(c_k)$ from the class name prompt c_k using the CLIP text encoder g. Then, pseudo-labels

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Algorithm 1 Training Process of GPS-CLIP

Input: Unlabeled training images $\mathcal{D}_u^{\text{tr}} = \{\mathbf{x}_i\}_{i=1}^{Q}$, class names $\{c_k\}_{k=1}^{K}$, and the fixed 106-template remote-sensing prompt ensemble released with RS-TransCLIP.
Parameters: Maximum outer rounds R; maximum inner iterations L; convergence threshold ϵ.
Output: Adapted and frozen CLIP model for unseen test inference.

for $r = 1$ **to** R **do**
  Extract features: $v_i = f(\mathbf{x}_i), \forall i$; $t_k = g(c_k), \forall k$.
  Initialize pseudo-labels $\{\hat{\mathbf{y}}_i\}_{i=1}^{Q}$ via Eq. (1).
  Compute affinity weights $w_{ij} = \max(0, v_i^\top v_j)$.
  Initialize $\boldsymbol{\mu}$, Σ, and set $\mathbf{z}_i^{(1)} = \hat{\mathbf{y}}_i$.
  while $|\mathcal{L}^{(l)} - \mathcal{L}^{(l-1)}| > \epsilon$ **do**
    for $l = 1$ **to** L **do**
      Update $\mathbf{z}_i^{(l+1)}$ for all $i \in Q$ via Eq. (7).
    end for
    Update μ_k and Σ via Eqs. (8)–(9).
    Compute total loss $\mathcal{L} = \mathcal{L}_{\text{GMM}} + \mathcal{L}_{\text{Lap}} + \mathcal{L}_{\text{KL}}$ via Eq. (6).
  end while
  Obtain refined probability matrix $\mathbf{Z}$.
  Select top-N_r samples per class as pseudo-labeled set $\mathbf{S}$.
  Fine-tune CLIP on $\mathbf{S}$ by minimizing $\mathcal{L}_{\text{CLIP}}$ via Eq. (14).
end for
Freeze the adapted model for inference on the test split.

$\{\hat{\mathbf{y}}_i\}$ are initialized according to (1), and the similarity-based weights $w_{ij} = \max(0, v_i^\top v_j)$ between image features are computed. Model parameters μ and Σ are initialized, and the latent labels are set as $\mathbf{z}_i^{(1)} = \hat{\mathbf{y}}_i$ for all i to begin the inner optimization loop.

During inner iterations, the latent labels and parameters are updated in the following manner: for each step l in the inner loop, we update the latent label $\mathbf{z}_i^{(l+1)}$ for all $i \in Q$ using (7). After L such updates, we compute the category mean μ_k and covariance matrix Σ using (8) and (9), and evaluate the current loss $\mathcal{L}$ via (6). The loop continues until $\mathcal{L}$ converges, yielding the final label matrix $\mathbf{Z}$.

After convergence, the top-N_r samples with highest confidence from $\mathbf{Z}$ are selected to form the pseudo-labeled set S, which is then used to retrain the CLIP model. Specifically, the model is updated by minimizing the CLIP contrastive loss $\mathcal{L}_{\text{CLIP}}$ defined in (14). This alternating process is repeated for R rounds using only $\mathcal{D}_u^{\text{tr}}$. The resulting model is then frozen and evaluated on the disjoint test split $\mathcal{D}^{\text{te}}$.

IV. Experiment and Analysis

A. Experiment Setup

We evaluate our GPS-CLIP on five standard remote sensing benchmarks: AID [41], EuroSAT [42], OPTIMAL31 [43], RESISC45 [5], and WHURS19 [44]. These datasets cover the vast majority of common remote sensing scene types, including urban, agricultural, forest, and water bodies, making them highly representative benchmarks. They enable a comprehensive evaluation of model performance across varying spatial resolutions, image scales, and class distributions.

GPS-CLIP follows an inductive train/test protocol. For each benchmark, the unlabeled training split $\mathcal{D}_u^{\text{tr}}$ used for adaptation and the test split $\mathcal{D}^{\text{te}}$ are disjoint. Pseudo-label generation, GMM fitting, graph construction, sample selection, and projection-layer optimization use only $\mathcal{D}_u^{\text{tr}}$. After adaptation, the model is frozen and evaluated on $\mathcal{D}^{\text{te}}$. No test image or test label is used in the adaptation process.

Text representations for pseudo-label generation and evaluation are obtained by averaging normalized embeddings from the fixed 106-template remote-sensing ensemble released with RS-TransCLIP [45] and used throughout our experiments. In the semi-supervised learning phase, we use the three-round schedule of 10, 20, and 30 pseudo-labeled training samples per class. We use Top-1 accuracy to measure ZSRSSC performance, which reflects the proportion of test samples for which the top-predicted class matches the ground truth label. During semi-supervised fine-tuning, we keep the backbone weights frozen and optimize only the CLIP image and text encoder projection layers. We use the AdamW optimizer with a learning rate of 1×10^{-5} and a batch size of 32, following a cosine annealing schedule with five warm-up steps per round. All main experiments run on a single NVIDIA A100 GPU with 80 GB memory. The random seed is fixed at 42. Adaptation takes roughly 3–8 min per dataset, depending on dataset size. The code and configuration files are publicly available at https://github.com/WXLuo13/GPS-CLIP

B. Ablation Studies

GPS-CLIP is designed to provide more representative, high-confidence pseudo-label samples for model training. In the first ablation experiment, we compare two settings: one with and one without the proposed pseudo-label refinement method. To make the comparison more intuitive, we evaluate the improvement in pseudo-label accuracy following refinement. Both settings use the same fixed 106-template prompt ensemble, so their difference isolates refinement rather than prompt design. Training-split labels are accessed only after predictions are fixed to compute this diagnostic accuracy; they are never supplied to GMM fitting, sample selection, or model updates.

Table I compares pseudo-label accuracy with and without GMM refinement under three sample sizes. With ViT-B/32, refinement improves the accuracy range from 95.9% to 98.1% to 98.4% to 99.5% on WHURS19. On OPTIMAL31, the corresponding range increases from 78.2% to 82.9% to 86.3% to 89.8%. These results demonstrate the effectiveness of GMM refinement across different sample sizes and backbones. Fig. 3 summarizes the comparison.

1) Prompt Prior and Method Contribution: The 106-template ensemble released with RS-TransCLIP is an inherited fixed remote-sensing prompt prior rather than a contribution of GPS-CLIP. To prevent prompt effects from being attributed to the proposed modules, every paired GPS-CLIP ablation uses the identical ensemble. Table I therefore isolates the effect of GMM refinement conditional on this fixed prompt representation. The progressive-schedule analysis below examines

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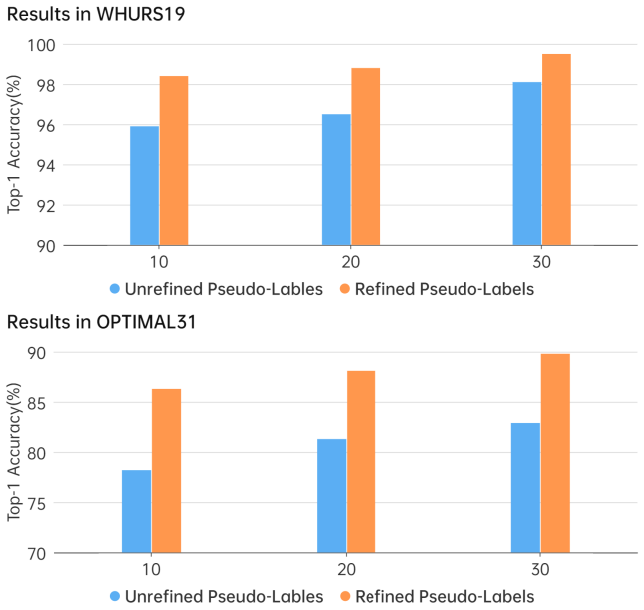


Fig. 3. Ablation study of pseudo-label refinement on WHURS19 and OPTIMAL31. Top-1 accuracy is shown for 10, 20, and 30 selected pseudo-label samples per class across three rounds, comparing results without and with our refinement module.

TABLE I

ABLATION STUDY OF PSEUDO-LABEL REFINEMENT ON WHURS19 AND OPTIMAL31. TOP-1 ACCURACY IS REPORTED FOR 10, 20, AND 30 SELECTED PSEUDO-LABEL SAMPLES PER CLASS ACROSS THREE ROUNDS. THE SYMBOLS × AND √ DENOTE RESULTS WITHOUT AND WITH OUR REFINEMENT MODULE, RESPECTIVELY. BOLD VALUES INDICATE THE HIGHEST ACCURACY FOR EACH BACKBONE AND DATASET

Backbone	Pseudo-Labels Refinement	WHURS19			OPTIMAL31		
		10	20	30	10	20	30
ResNet-50	×	96.8	97.9	99.0	81.0	82.7	83.3
	√	97.1	98.2	**99.2**	82.2	**84.8**	84.2
ViT-B/32	×	95.9	96.5	98.1	78.2	81.3	82.9
	√	98.4	98.8	**99.5**	86.3	88.1	**89.8**
ViT-L/14	×	96.1	97.1	97.9	90.4	91.6	93.0
	√	96.6	97.2	**98.5**	92.7	93.1	**94.2**

schedule sensitivity while holding both the prompt ensemble and refinement procedure fixed; it is not a separate with/without-progressive decomposition. We do not claim that gains caused by replacing a simple prompt with this ensemble originate from GPS-CLIP.

2) Effect of Unlabeled Data Quantity: GPS-CLIP relies on unlabeled training images for pseudo-label generation and GMM refinement. A natural question is how performance changes when only a fraction of the available unlabeled training split is used. We randomly subsample 20%, 40%, 60%, 80%, and 100% of $\mathcal{D}_u^{\text{tr}}$ on AID and RESISC45 with the ViT-B/32 backbone and the progressive sampling schedule. Table II reports frozen-model accuracy on the test split.

Even with only 20% of unlabeled data, GPS-CLIP already outperforms the no-adaptation CLIP baseline by 14.1 percentage points on AID and 10.6 percentage points on RESISC45. Performance grows steadily as more data becomes available,

TABLE II

EFFECT OF THE UNLABELED TRAINING-DATA PROPORTION ON TEST-SET TOP-1 ACCURACY. THE CLIP BASELINE WITHOUT ADAPTATION IS SHOWN FOR REFERENCE. ALL RESULTS USE THE VIT-B/32 BACKBONE

Unlabeled Ratio	AID	RESISC45
CLIP (no adapt)	66.4	60.7
20%	80.5	71.3
40%	83.7	74.6
60%	85.9	76.8
80%	87.4	78.5
100%	**88.1**	**79.5**

TABLE III

REPRESENTATIVE PER-CLASS PSEUDO-LABEL ACCURACY ON THE AID UNLABELED TRAINING SPLIT BEFORE AND AFTER GMM REFINEMENT, USING THE VIT-B/32 BACKBONE

Class	Without GMM	With GMM	Δ
DenseResidential	72.1	81.3	+9.2
MediumResidential	68.9	77.4	+8.5
Industrial	73.6	81.8	+8.2
Commercial	75.8	82.9	+7.1
Parking	79.4	85.7	+6.3
School	82.1	86.4	+4.3
Stadium	84.2	87.3	+3.1
River	88.3	90.1	+1.8
Forest	95.7	96.9	+1.2
Beach	97.8	97.9	+0.1

though the marginal gain slows down beyond 60%. From 20% to 60%, the accuracy improves by 5.4 points on AID and 5.5 points on RESISC45. From 60% to 100%, the corresponding gains are 2.2 and 2.7 points. The GMM refinement can therefore produce robust pseudo-labels from a moderate-sized unlabeled pool, making the method practical even when target-domain data is only partially available.

3) Class-Level Benefit of GMM Refinement: To understand which classes benefit most from the GMM refinement, we compare training-side pseudo-label accuracy before and after applying the GMM on AID with the ViT-B/32 backbone. This is a diagnostic: training labels are used only to score already fixed assignments and do not enter fitting or adaptation. Table III lists representative classes sorted by their accuracy gain Δ.

The largest gains come from classes whose visual features overlap heavily. DenseResidential and MediumResidential, for instance, share similar building layouts and color palettes, while industrial and commercial contain comparable structural elements. CLIP's initial cosine-similarity pseudo-labels are most confused on these pairs. The GMM resolves the ambiguity by leveraging cluster-level geometry together with Laplacian smoothing over the feature manifold. Highly distinctive classes like beach and forest already receive near-perfect initial pseudo-labels and therefore gain little from refinement.

4) Rationale and Limitation of the Gaussian Assumption: One Gaussian per class with a shared diagonal covariance is a low-variance approximation, not an assumption that remote-sensing categories are intrinsically unimodal. It preserves a direct correspondence between each component and

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TABLE IV

TOP-1 ACCURACY ON FIVE BENCHMARKS UNDER INDUCTIVE EVALUATION. GPS-CLIP ADAPTS ONLY ON THE UNLABELED TRAINING SPLIT AND IS EVALUATED WITH A FROZEN MODEL ON THE TEST SPLIT. METHODS THAT ADAPT ON TEST IMAGES ARE EXCLUDED. BOLD VALUES DENOTE THE BEST DISPLAYED RESULT FOR EACH BACKBONE AND DATASET

Backbone	Model	WHURS19	OPTIMAL31	EuroSAT	AID	RESISC45
ResNet-50	CLIP [19]	71.3	64.5	28.3	55.4	52.8
	RemoteCLIP [34]	95.4	64.0	26.7	89.1	51.6
	GPS-CLIP (ours)	**98.8**	**83.6**	**35.5**	**92.1**	**73.4**
ViT-B/32	CLIP [19]	81.1	73.0	45.3	66.4	60.7
	GeoRSCLIP [35]	88.8	79.6	**53.4**	70.3	68.8
	RemoteCLIP [34]	95.2	77.6	35.5	**91.7**	68.1
	SkyCLIP [47]	91.0	79.5	52.6	70.3	66.7
	GPS-CLIP (ours)	**95.6**	**83.7**	53.2	88.1	**79.5**
ViT-L/14	CLIP [19]	85.5	80.6	60.1	69.7	71.3
	GeoRSCLIP [35]	88.5	83.7	59.9	74.4	73.8
	RemoteCLIP [34]	93.5	83.8	43.6	84.1	76.0
	SkyCLIP [47]	87.5	80.9	51.5	72.1	70.5
	GPS-CLIP (ours)	**96.7**	**93.8**	**74.1**	**92.4**	**85.5**

TABLE V

TOP-1 ACCURACY FOR DIFFERENT NUMBERS OF PSEUDO-LABEL SAMPLES. PSEUDO ACC. IS THE TOP-1 ACCURACY OF THE PSEUDO-LABELS AND TEST ACC. IS THE TOP-1 ACCURACY ON THE TEST SET. Δ INDICATES THE ACCURACY CHANGE RELATIVE TO THE PREVIOUS ITERATION. BOLD Δ VALUES DENOTE A DECREASE

Samples	WHURS19		AID	
	Pseudo Acc.	Test Acc.	Pseudo Acc.	Test Acc.
10	98.4	90.8	85.1	79.2
20	98.8 $\Delta = +0.4$	92.5 $\Delta = +1.7$	88.3 $\Delta = +3.2$	84.4 $\Delta = +5.2$
30	99.5 $\Delta = +0.7$	95.6 $\Delta = +3.1$	90.2 $\Delta = +1.9$	88.1 $\Delta = +3.7$
40	99.6 $\Delta = +0.1$	95.2 $\mathbf{\Delta = -0.4}$	92.7$\Delta = +2.5$	88.4 $\Delta = +0.3$
50	99.5 $\mathbf{\Delta = -0.1}$	95.0 $\mathbf{\Delta = -0.2}$	92.9 $\Delta = +0.2$	87.7 $\mathbf{\Delta = -0.7}$

a semantic class and avoids unstable class-specific covariance estimates from noisy pseudo-labels. The tradeoff is that region-, resolution-, sensor-, and acquisition-dependent subclusters cannot be represented explicitly. This limitation can affect complex classes, such as residential, industrial, and commercial scenes. We therefore retain the stable single-component design and leave training-only mixture selection and adaptive component numbers to future work.

C. Main Results

Table IV compares methods under inductive evaluation and excludes methods that adapt on the test batch. GPS-CLIP obtains the best displayed result in 13 of the 15 backbone–dataset settings: all five settings with ResNet-50, three with ViT-B/32, and all five with ViT-L/14. We therefore characterize the method as superior or competitive in most settings rather than uniformly state of the art.

The two nonbest settings are both obtained with ViT-B/32. On EuroSAT, GPS-CLIP reaches 53.2%, close to GeoRSCLIP at 53.4%; the 0.2-point gap may reflect the sensitivity of the small ten-class dataset to prompt and pretraining priors. On AID, GPS-CLIP reaches 88.1% versus 91.7% for RemoteCLIP. AID has high intraclass diversity, which can bias a single-component estimate and preserve early pseudo-label errors. These are plausible factors rather than controlled causal conclusions.

D. Sensitivity Analysis

The main configuration uses three progressive rounds with 10, 20, and 30 pseudo-labeled training samples per class, with $L = 5$ and $M = 8$. We further report sensitivity to these settings below.

Table V reports five iterations on WHURS19 and AID with ViT-B/32. While training-side pseudo-label accuracy generally increases, test accuracy does not always follow this trend. On WHURS19, test accuracy drops by 0.4 and 0.2 percentage points in the fourth and fifth iterations, respectively. On AID, it decreases by 0.7 percentage points in the fifth iteration. This behavior is consistent with overfitting to pseudo-label noise and motivates training-side stopping criteria in future work.

We use RESISC45 and WHURS19 to study the effect of pseudo-label sample size. In this sensitivity analysis, the four schedules are evaluated under the same fixed configuration so that only the sample size changes. Fig. 4 compares their accuracy over three iterations. The schedule that selects 10, 20, and 30 samples per class achieves the highest third-round accuracy on both datasets. Increasing the sample size further introduces lower confidence labels and does not consistently improve performance.

1) Analysis of Iteration Degradation and Mitigation Strategies: As shown in Table V, test accuracy on WHURS19 drops by 0.4 and 0.2 percentage points in the fourth and fifth iterations, respectively. On AID, it drops by 0.7 percentage points in the fifth iteration even though pseudo-label accuracy continues to rise. Two factors explain this divergence. First, as N_r grows, later rounds incorporate lower confidence samples whose labeling errors compound across iterations. Second, CLIP's large capacity relative to only 30–50 samples per class makes the model prone to fitting pseudo-label noise instead of learning the underlying class semantics.

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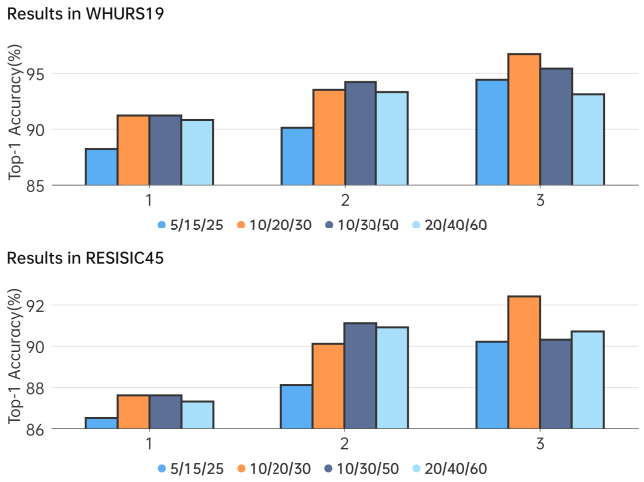


Fig. 4. Accuracy trends over three iterations for four pseudo-label sampling schedules on WHURS19 and RESISC45.

TABLE VI

SENSITIVITY TO INNER ITERATION COUNT L AND GMM INITIALIZATION SIZE M ON WHURS19 AND AID. ALL RESULTS USE THE VIT-B/32 BACKBONE WITH THE PROGRESSIVE SAMPLING SCHEDULE. BOLD VALUES INDICATE THE MAIN CONFIGURATION

Inner iteration L (fix $M=8$)			Init. size M (fix $L=5$)		
L	WHURS19	AID	M	WHURS19	AID
1	94.2	85.8	2	94.8	86.9
3	95.0	87.2	4	95.2	87.4
5	**95.6**	**88.1**	8	**95.6**	**88.1**
10	95.4	87.9	16	95.3	87.7
20	95.1	87.5	32	94.7	87.1

Several strategies can mitigate this degradation. Early stopping can halt training when the reduction in pseudo-label entropy falls below a threshold δ. Adaptive sample selection can add new samples only when the round-level confidence $c_r = 1/|Q| \sum_{i \in Q} \max_k \mathbf{Z}_{i,k}$ exceeds a percentile threshold. Confidence-based filtering can also require $\max_k \mathbf{z}_{i,k} \geq \rho$ in addition to the Top-N_r criterion, where ρ is set to 0.7. We leave a systematic comparison of these strategies to future work. The varying post hoc gains indicate that a training-side confidence-aware stopping rule may be more suitable when the round-zero assignments are already reliable.

2) GMM Component Number K: The number of Gaussian components K equals the number of target classes, which preserves a direct correspondence between components and classes. This parsimonious choice reduces estimation variance under noisy training-side pseudo-labels, but it cannot explicitly represent multimodal substructure within a semantic class.

3) Inner Iteration Count L and GMM Initialization M: We also study sensitivity to two additional hyperparameters: the maximum number of inner GMM refinement iterations L, and the number of most confident samples M used to initialize each class mean μ_k. Table VI reports Top-1 accuracy on WHURS19 and AID using the ViT-B/32 backbone with the progressive sampling schedule.

Performance remains stable across a wide range of both L and M. When L ranges from 1 to 20, accuracy spans 1.4 points on WHURS19 and 2.3 points on AID. When M ranges from 2 to 32, it spans 0.9 and 1.2 points, respectively. The method therefore does not require precise tuning. The main values $L = 5$ and $M = 8$ lie near the center of the stable region and obtain the highest values in this table. Too few inner iterations leave the assignment under-refined, while too many iterations or an excessively large initializer set tends to dilute the per-class distribution estimate.

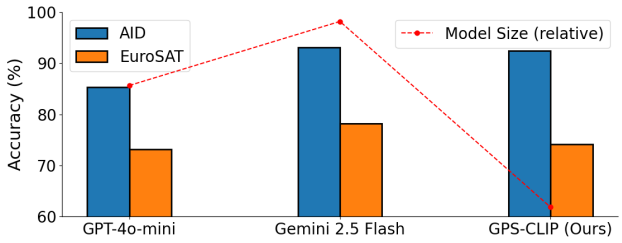


Fig. 5. Top-1 accuracy of GPS-CLIP with ViT-L/14 versus GPT-4o-mini and Gemini 2.5 Flash, shown as a supplementary reference.

TABLE VII

TOP-1 ACCURACY OF GPS-CLIP WITH VIT-L/14 VERSUS GPT-4O-MINI AND GEMINI 2.5 FLASH. THIS COMPARISON SERVES AS A SUPPLEMENTARY REFERENCE. DUE TO DIFFERENCES IN INFERENCE PROTOCOLS, THE RESULTS SHOULD BE INTERPRETED WITH CAUTION

Model	AID	EuroSAT
GPT-4o-mini	85.3	73.1
Gemini 2.5 Flash	93.1	78.2
GPS-CLIP (Ours, ViT-L/14)	92.4	74.1

E. Comparison With Commercial VLMs

To put GPS-CLIP in a broader practical context, we compare it with two commercial VLMs: GPT-4o-mini and Gemini 2.5 Flash. This comparison is intended as an informal reference rather than a controlled benchmark. Commercial models differ fundamentally from CLIP-based methods in architecture, training data, and inference protocol, so the results should be interpreted with care.

We evaluate on the AID and EuroSAT datasets. For the two commercial models, we use a forced-choice prompting strategy with the following fixed template: *"This is a remote sensing image. Please classify it into exactly one of the following categories: [class$_1$], [class$_2$],..., [class$_K$]. Output only the category name, with no additional explanation."* Each image receives a single response with no rejection handling or multitrial averaging. GPS-CLIP follows its standard zero-shot inference pipeline with the ViT-L/14 backbone. The two evaluation protocols are not equivalent. GPS-CLIP performs embedding-space nearest-neighbor matching, whereas the commercial models generate text responses under a generative forced-choice constraint. We report Top-1 accuracy for all models.

As shown in Table VII and Fig. 5, GPS-CLIP reaches 92.4% on AID and 74.1% on EuroSAT, surpassing GPT-4o-mini on both datasets. Gemini 2.5 Flash scores higher at 93.1% on AID and 78.2% on EuroSAT, which is expected given its large-scale proprietary training data. Still, the gap between

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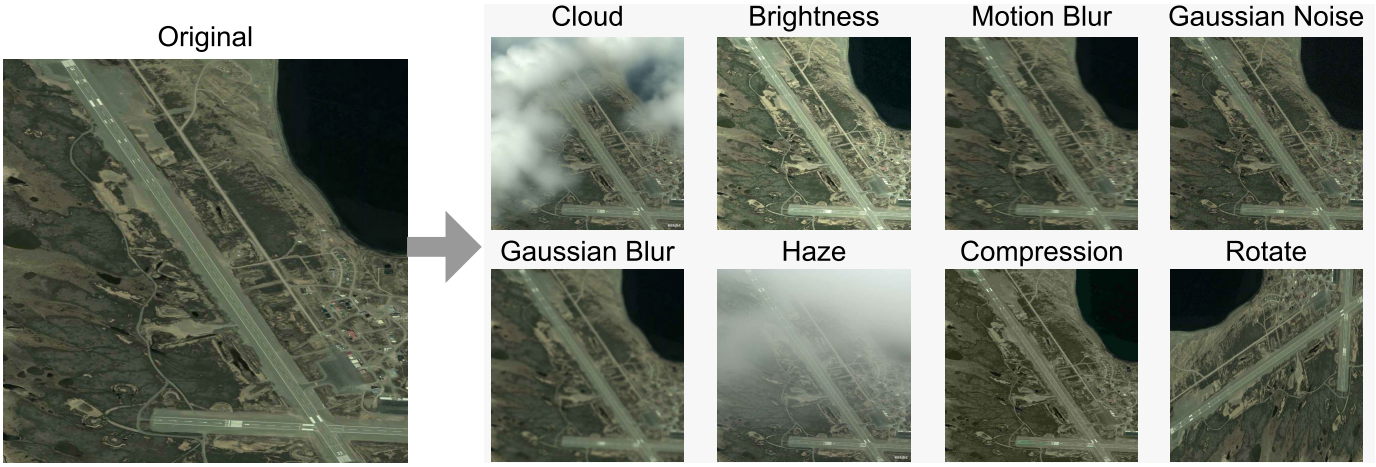


Fig. 6. Examples of comparisons between the original image and eight perturbed images.

TABLE VIII

TOP-1 ACCURACY AND Δ_{TP} OF MODELS ON OPTIMAL31: CLOUD, BRIGHTNESS VARIATION, MOTION BLUR, GAUSSIAN NOISE, GAUSSIAN BLUR, HAZE, COMPRESSION ARTIFACTS, AND ROTATION. THE BEST RESULT IS IN BOLD

Datasets	Model	Original	Cloud	Brightness	Motion Blur	Gaussian Noise	Gaussian Blur	Haze	Compression	Rotate	Δ_{TP}
OPTIMAL31	CLIP	73.0	70.2	70.9	49.3	66.4	65.9	71.2	67.7	73.0	6.2
	RemoteCLIP	77.6	75.1	76.2	56.8	70.8	73.9	75.6	71.4	77.4	5.5
	GPS-CLIP (ours)	**83.7**	**80.5**	**81.2**	**65.4**	**79.0**	**80.6**	**82.1**	**79.5**	**83.5**	**4.7**

GPS-CLIP and Gemini 2.5 Flash on AID is only 0.7%. GPS-CLIP is built on a publicly available ViT-L/14 backbone and achieves competitive performance through targeted remote sensing domain adaptation rather than sheer model scale. This suggests a favorable tradeoff between accuracy and accessibility for remote sensing applications. Because the evaluation protocols differ fundamentally, we refrain from drawing strong quantitative conclusions from this comparison.

F. Robustness Experiment

To validate the robustness of GPS-CLIP, we conducted experiments to demonstrate its stability when processing perturbed images. For this purpose, we introduced eight types of corruptions [47] to the OPTIMAL31 dataset. As illustrated in Fig. 6, these corruption types include cloud, brightness variation, motion blur, gaussian noise, gaussian blur, haze, compression artifacts, and rotation.

Comparative models include CLIP and RemoteCLIP. All use the same setup as the main experiments. Corruptions are applied to the test split after GPS-CLIP adaptation is complete, with the model kept frozen. We use Top-1 accuracy as the evaluation metric. To further quantify the corruption robustness of models, we introduce the task performance drop (Δ_{TP}) [47] following the definition in REOBench. Δ_{TP} is defined as

$$\Delta_{TP} = \mathbb{P}_{(x,y)\sim\mathcal{D}}\left[f(x) = y\right] - \mathbb{E}_{c\sim\mathcal{C}}\left[\mathbb{P}_{(x,y)\sim\mathcal{D}}\left[f(c(x)) = y\right]\right] \tag{15}$$

where $\mathbb{P}_{(x,y)\sim\mathcal{D}}[f(x) = y]$ denotes the model's accuracy on original images, and $\mathbb{E}_{c\sim\mathcal{C}}[\mathbb{P}_{(x,y)\sim\mathcal{D}}[f(c(x)) = y]]$ represents the average accuracy across all corrupted scenarios. A smaller Δ_{TP} indicates stronger robustness against image quality degradation.

As shown in Table VIII, we analyze the robustness of GPS-CLIP and two comparative models on the OPTIMAL31 dataset under eight types of image corruptions, with Top-1 accuracy and task Δ_{TP} as core metrics for evaluating robustness. From the perspective of overall robustness, GPS-CLIP achieves the smallest Δ_{TP} of 4.7% among all models, which is lower than CLIP 6.2% and RemoteCLIP 5.5%. These results indicate that GPS-CLIP has the lowest average performance drop under the reported corruption protocol.

GPS-CLIP performs particularly well under motion blur, achieving 65.4% compared with 49.3% for CLIP and 56.8% for RemoteCLIP. It also maintains stable performance under cloud, brightness variation, and rotation. Accuracy reaches 80.5% under cloud and 83.5% under rotation, compared with 83.7% on clean images. These results demonstrate the robustness of GPS-CLIP under different image degradations.

G. Cross-Dataset Generalization

To assess how well the learned representations generalize when the adaptation data comes from a different distribution than the evaluation data, we conduct a cross-dataset experiment with the ViT-B/32 backbone. We adapt GPS-CLIP on the unlabeled images of one source dataset and then evaluate the frozen model on another dataset using only its own class names. We test two transfer directions between AID and RESISC45. These two datasets contain 30 and 45 classes, respectively, and share a substantial number of semantically overlapping categories, such as airport, beach, forest, parking, river, and stadium.

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TABLE IX

CROSS-DATASET GENERALIZATION RESULTS WITH THE VIT-B/32 BACKBONE. "ADAPT" DENOTES THE DATASET USED FOR UNLABELED ADAPTATION AND "EVAL" DENOTES THE DATASET USED FOR ZERO-SHOT EVALUATION

Setting	AID	RESISC45
CLIP (no adaptation)	66.4	60.7
GeoRSCLIP (no adaptation)	70.3	68.8
GPS-CLIP (adapt on AID, eval on RESISC45)	—	68.2
GPS-CLIP (adapt on RESISC45, eval on AID)	73.8	—
GPS-CLIP (in-domain reference)	88.1	79.5

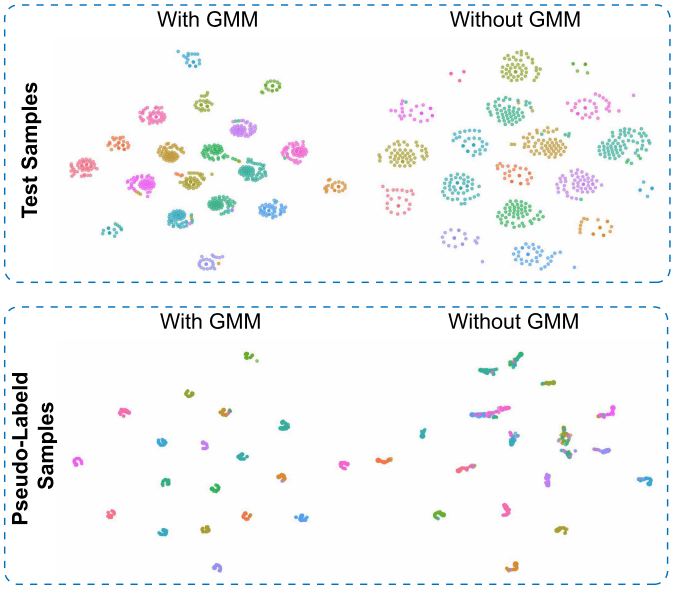


Fig. 7. T-SNE visualizations of postfreeze test samples and selected training-side pseudo-labeled samples on WHURS19 for GPS-CLIP and the model without GMM refinement.

As shown in Table IX, cross-dataset adaptation outperforms the no-adaptation CLIP baseline on both evaluation sets, improving by 7.5 percentage points on RESISC45 when adapting on AID and by 7.4 percentage points on AID when adapting on RESISC45. This confirms that the GMM-refined features and contrastive fine-tuning transfer general remote sensing knowledge across datasets. Cross-domain performance is roughly on par with GeoRSCLIP, which requires dedicated remote sensing pretraining but no adaptation. A gap remains relative to in-domain GPS-CLIP. This is expected because the GMM cluster means μ_k are estimated from source-domain features and may not perfectly align with the target-domain feature manifold when class distributions shift. We acknowledge this limitation and consider approaches like optimal transport alignment and domain-adversarial GMM initialization as promising directions for future work.

H. Visual Analysis

To further explore the impact of the GMM-based pseudo-label refinement module on feature distribution and class separability, we conduct a visualization analysis using t-SNE [48]. The experiment employs the WHURS19 dataset and adopts the ViT-B/32 backbone, which is consistent with the main experiment to ensure the validity of the result. We specifically design an ablation study on the GMM module: the full GPS-CLIP model and the ablated model that only retains initial pseudo-labels from CLIP without GMM refinement. Our analysis focuses on comparing feature plots of image samples and selected pseudo-labeled samples.

Fig. 7 compares selected training-side pseudo-labeled features and postfreeze test features for GPS-CLIP and the model without GMM refinement. GPS-CLIP produces tighter intraclass clusters and clearer interclass boundaries than the model without GMM refinement. Some classes still show dispersed or overlapping clusters, indicating that one Gaussian may not fully describe all class distributions.

V. CONCLUSION

This article proposes GPS-CLIP for inductive ZSRSSC. GPS-CLIP refines CLIP pseudo-labels on an unlabeled training split by jointly modeling class-conditional density, neighborhood relations, and semantic consistency. A class-balanced progressive strategy then selects high-confidence training samples to adapt the projection layers. The adapted model is frozen for inference on a disjoint test split. Experiments on five datasets and three backbones show superior or competitive performance among inductive methods in most settings.

The shared single-Gaussian model provides stable estimates, but it may not fully capture multimodal classes across regions, sensors, spatial resolutions, and acquisition conditions. Cross-dataset performance also remains lower than in-domain adaptation. Future work will investigate training-only adaptive component selection, confidence-aware stopping, and distribution alignment for more robust cross-domain adaptation.

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